

```
import cv2
import numpy as np
from google.colab import files
from matplotlib import pyplot as plt

# Upload the image
uploaded = files.upload()

# Get uploaded image name
filename = list(uploaded.keys())[0]

# Read image in grayscale
img = cv2.imread(filename, 0)

# Check whether image is loaded
if img is None:
    print("Error: Image could not be loaded.")
else:
    # Sobel X
    sobelx = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3)

    # Sobel Y
    sobely = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3)

    # Combine Sobel X and Sobel Y
    edges = cv2.addWeighted(sobelx, 0.5, sobely, 0.5, 0)

    # Convert to 8-bit image
    edges = cv2.convertScaleAbs(edges)

    # Save output
    cv2.imwrite("Edge_detection.jpg", edges)

    # Display images
    plt.figure(figsize=(10, 4))

    plt.subplot(1, 2, 1)
```

```
plt.imshow(img, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(edges, cmap="gray")
plt.title("Sobel Edge Detection")
plt.axis("off")

plt.show()

# Download output
files.download("Edge_detection.jpg")
```

cvvv8.png

cvvv8.png(image/png) - 773706 bytes, last modified: 7/22/2026 - 100% done

Saving cvvv8.png to cvvv8.png

Original Image



Sobel Edge Detection



